COURSE PLAN

Introduction to Data Structures and Algorithms (ES 248)

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Course Overview

This course is a bridge between Computing (in Python) and DSA 1 (in C).

Course positioning:

ES 112/113 (Python) → This Course (C + Basic DSA) → DSA 1 (DSA in C/C++)

If you've learned Python and wondered "what's really happening when my code runs?", this course reveals what Python abstracts away: memory allocation, pointers, and how data structures actually work in hardware.

Prerequisites

- Introduction to Programming
- Comfortable with programming basics (variables, loops, functions, lists)

Who Should Take This Course

- Students planning to take DSA 1 or systems-level courses
- Anyone interested in systems programming, embedded systems, IoT
- Students preparing for technical interviews (many require C/systems knowledge)
- Those curious about low-level programming and computer architecture

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Learning Outcomes

By the end of this course, you will:

1. Write, compile, and debug C programs
2. Understand memory management: stack vs heap, pointers, dynamic allocation
3. Implement fundamental data structures (arrays, linked lists, stacks, queues) in C
4. Analyze time and space complexity
5. Work with file I/O and data processing at systems level
6. Understand performance implications of different approaches
7. Be prepared for advanced DSA course in C

Topics Covered

- Linux basics: Command line, file system, compilation workflow
- Python to C transition: Compilation, types, memory model
- Arrays and memory: How arrays really work, operations and costs
- Pointers : Introduction, arithmetic, dynamic memory (malloc/free)
- Structures and Unions: Custom data types, memory layout, grouping related data
- Data structures: Linked lists, stacks, queues (array vs linked implementations)
- Algorithms: Searching, sorting, complexity analysis
- File I/O: Reading/writing files, parsing CSV/TSV data
- Systems concepts: Bit manipulation, memory layout

Resources

Primary: Charles Severance, "C Programming for Everybody" (Free online: https://www.cc4e.com/)

Video Resources**:** MyCodeSchool C and Data Structures playlist (https://www.youtube.com/@mycodeschool)

References: K&R "The C Programming Language", K.N. King "C Programming: A Modern Approach", Michael Barr and Andy Oram. 1998. “Programming Embedded Systems in C and C++” (1st. ed.). O'Reilly & Associates, Inc. USA., Yedidyah Langsam, Moshe J. Augenstein, and Aaron M. Tenenbaum. 1996. Data structures using C and C++ (2nd ed.).

Assessment

The grading scheme will be announced in the first week.

Assessment will include a combination of:

- Lab assignments and coding exercises
- Quizzes
- Mid-semester exam
- Final exam

Teaching Format

- Lectures: Concepts, demonstrations, Python vs C comparisons
- Labs: Hands-on coding, implementation exercises
- Approach: May include flipped elements (videos before class, coding during class)

Development tools: C compiler, text editor/IDE, debugger (setup help provided in first lab)

Questions?

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Detailed syllabus, schedule, and policies will be shared in the first week of class.